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Latent Profiles of Intrinsic Interpersonal Emotion Regulation in Chinese Children: Links to Psychological Adjustment

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Children use interpersonal emotion regulation (IER) with parents and friends to manage their emotions, which impacts their psychological adjustment. However, the distinct patterns of IER with parents and friends and their effects on children's psychological adjustment are not well understood. The present study employs latent profile analysis to identify unique patterns of intrinsic IER with parents and friends among a cohort of Chinese children (N = 1,678, M_{age} = 10.42 years; SD = 1.2 years; 50.9% boys) and explores the associations of those subgroups with children's psychological adjustment (including depressive symptoms, anxiety symptoms, general self-worth, and life satisfaction). The findings revealed six distinct intrinsic IER profiles: extremely low IER (2.5%), low IER (16.2%), average IER (40.2%), high parent-low friend IER (4.2%), low parent-high friend IER (3.9%), and high IER (33.0%). Children in higher grades and girls were more likely to belong to the low parent-high friend IER profile compared to their counterparts. Children in high IER with both parents and friends reported the best psychological adjustment. In contrast, children categorized in the extremely low IER, low IER, and low parent-high friend IER profiles displayed poorer psychological adjustment relative to average IER and high IER profiles. These findings highlight the importance of examining the diversity of intrinsic IER patterns with parents and friends to gain a comprehensive understanding of children's psychological adjustment.

Keywords: interpersonal emotion regulation, psychological adjustment, children, latent profile analysis

Interpersonal emotion regulation (IER) refers to a goal-directed process by which individuals influence their own or others' emotional states through social interactions (e.g., Dixon-Gordon, Bernecker, & Christensen, 2015; Niven, 2017; Zaki & Williams, 2013). IER encompasses two types of processes: *intrinsic* IER, which involves seeking help to regulate one's own emotions, and *extrinsic* IER, which refers to influencing another person's emotions through regulatory efforts (Zaki & Williams, 2013). Compared to adults, children exhibit limited self-regulation capacities and therefore often rely on external sources such as parents, peers, and teachers to manage their emotions (Gee et al., 2014; Steinberg et al., 2018). Given this developmental dependency, the present study centers on intrinsic IER, wherein children seek regulatory support from others. Previous studies have demonstrated that intrinsic IER plays a crucial role in enhancing children's emotional well-being (e.g., Ding et al., 2021; Do et al., 2025; Stone et al., 2019). While existing studies have highlighted the importance of the characteristics of regulators (i.e., the person from whom such support is sought) in shaping IER outcomes (e.g., Niven, 2022; Springstein et al., 2025; Thompson et al., 2024), limited attention has been paid to the distinct roles played by different types of regulators in influencing children's psychological adjustment.

Both parents and friends serve as key regulators in handling children's emotions (e.g., Eisenberg, 2020; Kovacs et al., 2006; R. L. Miller-Slough & Dunsmore, 2016). However, their influence on children's emotional development and adjustment may differ. Prior research using a variable-centered approach has examined how

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frequently parents and friends use specific strategies or respond to children's emotions, revealing that they may differentially affect children's emotion regulation abilities (e.g., Hale et al., 2023; R. Miller-Slough & Dunsmore, 2019; J. Wang, Wang, et al., 2024) and depressive symptoms (Do et al., 2025).

It is important to note that these differential effects between parents and friends may also stem from individual differences in children's tendency to seek IER and their perceptions of its effectiveness with a given regulator (i.e., parents or friends). To date, however, little research has explored the heterogeneity of IER patterns—specifically, how children seek emotion regulation support from different regulators and how these engagement patterns relate to their psychological adjustment. Thus, the present study focused on children's tendency to seek intrinsic IER and their perception of its efficacy in managing their emotions. Using a person-centered approach, it aimed to identify distinct intrinsic IER patterns and examine how these patterns are related to children's psychological adjustment.

IER With Different Regulators Among Children

Intrinsic IER can function effectively through the process of seeking emotional support from others (e.g., sharing emotions with others), regardless of how regulators actually respond, highlighting the response-independent characteristic of IER processes (Zaki & Williams, 2013). Importantly, the extent to which individuals seek IER varies depending on the type of relationship. Prior studies have shown that people tend to share emotions with specific partners rather than uniformly across all relationships (e.g., D. Y. Liu et al., 2021; Ruan et al., 2020). Cheung et al. (2015) proposed that individuals strategically distribute their emotion regulation needs across multiple specialized relationships, rather than relying on a single person for all emotion support—for example, turning to a sister for regulating anger but to a best friend for regulating sadness. In other words, individuals would rely on different regulators to satisfy their different emotion regulation needs.

Diversification in seeking emotional support across relationships is particularly relevant for children, given their developing capacity for self-regulation (Zachariou & Whitebread, 2019). As children grow older, they increasingly engage in IER across multiple relationships, resulting in varying degrees of reliance on others for emotion regulation (Barthel et al., 2018; Kwon et al., 2025). Parents typically serve as primary regulators in early childhood, whereas peers gradually assume a more prominent role as children mature (Bowlby, 1982; Gorrese & Ruggieri, 2012). Accordingly, the present study examined parents and friends as key regulators during middle and late childhood, as children may turn to each to fulfill distinct emotion regulation needs.

IER With Parents and Friends and Children's Psychological Adjustment

Parents play a pivotal role in children's social and emotional development during childhood and adolescence (Barthel et al., 2018; Eisenberg, 2020; Morris et al., 2007). Due to their greater emotional maturity and regulatory competence, parents are often more effective than children themselves in regulating emotions, serving as emotional coaches throughout the IER process (von Salisch, 2001). Notably, research has shown that even in the mere presence of their mother without direct interaction, children were found to exhibit improved regulatory behavior in the laboratory setting (Gee et al., 2014). As primary attachment figures, parents thus serve as crucial external regulators who help children manage their emotions, ultimately promoting their better mental health and emotional well-being. As children age, their social network expands beyond family, and peers' role becomes salient in children's IER process. Research among adolescents has shown that greater perceived emotional support from friends is associated with lower average levels of daily negative affect (Cui et al., 2020; H. Wang, Xu, et al., 2024), and that more frequent affective expression with friends is linked to fewer internalizing problems (Lindsey, 2021). Unlike the top-down nature of parent–child relationships, friendships are typically characterized by a more symmetrical, "horizontal" structure (von Salisch, 2001), in which children are more likely to disclose their emotions freely to maintain social bonds. Nonetheless, this growing reliance on friends is not unequivocally beneficial. For example, excessive self-disclosure with friends can lead to corumination, which is often associated with greater internalizing problems (Rose, 2002, 2021). It suggests that friends may not always provide effective support due to their limited cognitive and emotional capacities, which can undermine the success of IER (Niven et al., 2024). Critically, although parents and friends serve as distinct regulators, children's social and emotional development is shaped by the dynamic interplay between these relationships (e.g., Collins et al., 2000; Hu et al., 2021). A growing body of research has examined the simultaneous roles of parents and friends in the IER process and compared their effects, but the results have been mixed. For instance, Do et al. (2025) used ecological momentary assessment to examine the IER strategies employed by parents and friends in adolescent girls' daily lives. Their results showed that while engaging in IER with either parents or friends significantly predicted a decreased likelihood of experiencing negative affect, only a higher frequency of adaptive IER strategies used by parents was related to fewer depressive symptoms 1 year later (Do et al., 2025). In contrast, some studies have found that friends' responses exert a greater influence than parents on adolescents' emotion regulation ability (Hale et al., 2023; J. Wang, Wang, et al., 2024). Moreover, Stone et al. (2019) explored whether the source of support (i.e., parents vs. peers) moderated the effectiveness of social regulation strategies (e.g., coproblem solving) in reducing negative affect among youth with clinical anxiety and found no significant results, indicating no differential effects between parents and friends.

Previous studies have primarily focused on the strategies employed by regulators, overlooking children's individual differences in their tendency to seek IER and their perceived efficacy of IER with parents and friends. Specifically, children's tendency to engage in IER may determine whether they seek and accept support from regulators, and their perceived efficacy of IER may influence how effective regulators' strategies are (Lemay et al., 2025; Williams et al., 2018). Therefore, to better understand the differential roles of parents and friends in children's psychological adjustment, it is important to consider both the tendency and perceived efficacy of IER.

A Person-Oriented Approach to IER With Different Regulators

Previous studies on IER and psychological adjustment have predominantly utilized a variable-centered approach at the individual level (e.g., Do et al., 2025; Gadassi-Polack et al., 2025; Kwon & Lopez-Perez, 2022; Stone et al., 2019). This approach assumes homogeneity within the population and explores associations between variables of interest while overlooking the potential variations in the associations between IER and psychological adjustment across subgroups (e.g., Howard & Hoffman, 2018). In contrast, the person-centered approach could reveal the heterogeneity among the people and was mainly utilized to reveal different subgroups with a particular collection of emotion regulation strategies endorsed, especially for intrapersonal emotion regulation (e.g., Dixon-Gordon, Aldao, & De Los Reyes, 2015; Grommisch et al., 2020; Li et al., 2025). However, in addition to a broad range of strategies, as mentioned above, children would seek different regulators for help to handle emotions as they age and achieve their regulatory goals, potentially resulting in diverse outcomes (Barthel et al., 2018). During the process, the reliance on different regulators may coexist among children, and not all children apply the same level of IER across various relationships as well. Thus, instead of using variable-centered approaches about different regulators, the person-centered approach may better elucidate the heterogeneous subgroups within the IER process across different regulators and explore the characteristics of these subgroups, enabling more targeted intervention.

To the best of our knowledge, few studies have examined individuals' tendencies to use IER and their evaluations of its efficacy with different regulators using the person-centered approach. However, in related fields such as attachment and social support research, which are closely linked to the construct of IER (e.g., Dixon-Gordon, Bernecker, & Christensen, 2015; Kappas, 2013), prior studies have investigated the corresponding subgroups across various relationships. For instance, Laible et al. (2000) examined the relations of parent and peer attachment to adolescents' adjustment ($M_{age} = 16.1$ years, $SD = 1.8$) and identified four groups based on their parent and peer attachment scores: (a) high parent–high peer attachment group, (b) high peer–low parent attachment group, (c) high parent–low peer attachment group, and (d) low parent–low peer attachment group. Regarding the social support profiles, previous studies investigated profiles of perceived social support among adolescents from multiple sources (e.g., classmates, teachers, family, and friends; Ciarrochi et al., 2017; Laursen et al., 2006; Petersen et al., 2023; Scholte et al., 2001) and finally identified different levels of social support profiles from all sources (e.g., "high support," "moderate support," and "low support"), as well as divergent or mixed profiles characterized by the high levels of social support from one or two sources while low levels of social support from other sources (e.g., "low friend support," "parent-peer support"). These findings suggest that mixed support profiles are both theoretically and empirically plausible in the IER context. Thus, it is reasonable to infer that there may be different patterns characterized by both high and low levels of IER with parents and friends, as well as mixed patterns characterized by high levels of IER with only one source.

Regarding the association between IER patterns with different regulators and children's psychological adjustment, having a wider range of regulators available can provide more opportunities to accomplish individuals' regulatory goals and help them regulate emotions effectively based on the relational regulation theory (Lakey & Orehek, 2011). Thus, children who rely heavily on both parents and friends may exhibit better psychological adjustment. However, it remains uncertain whether children who rely less on friends for regulating their emotions can buffer or compensate for this by relying heavily on parents. Conversely, it is also unclear whether friends can act as a buffer against maladjustment in children who seek less help from parents for emotion regulation. These research questions require further investigation to better understand the IER patterns with different regulators and their impact on children's psychological adjustment.

The Role of Grade and Gender

Among children, there are also age-related (i.e., grade) and gender differences in the tendency to rely on parents or friends as regulators for emotion regulation. According to the proximity-seeking theory, children gradually spend more time with their friends, and physical closeness may facilitate the formation of attachment (Hazan & Shaver, 1994). This increased proximity provides children with more opportunities to seek help or share their feelings with friends. For instance, as children grow up and progress to a higher grade, they are more likely to turn to best friends rather than parents for proximity-seeking and safe haven functions (Nickerson & Nagle, 2005). It was found that fourth graders rated their mothers and fathers as the most frequent providers of support, while the same-sex friend was rated as highest in the seventh grade (Furman & Buhrmester, 1992). As such, it would be reasonable to posit that children in the high grade are more inclined to turn to friends for emotional support, as friends are the greatest source of companionship (Furman & Buhrmester, 1985).

Previous research has indicated significant gender disparities in peer relationship dynamics, with girls exhibiting higher levels of self-disclosure in friendships during middle childhood and early adolescence compared to boys (e.g., Rose, 2002; Rose & Rudolph, 2006; Valkenburg et al., 2011). Specifically, it was found that third-, fourth-, and fifth-grade girls were more inclined to share their feelings with friends than boys (Kwon & Lopez-Perez, 2022; Rose, 2002), as well as preadolescent girls (ages range 10–11; Valkenburg et al., 2011). Moreover, 9th- to 12th-grade girls are more likely to seek support from female friends (Sears & McAfee, 2017). Another two studies also revealed that preadolescent girls nominated more peers to regulate their emotions than did boys (Kwon et al., 2022, 2025). Thus, it is reasonable to expect that girls will be more active when seeking others' help to regulate their own emotions, especially among friends. Altogether, to gain a more nuanced picture of IER with different regulators, the study also explores the role of grade and gender on IER patterns.

The Present Study

It is important to acknowledge that existing research has primarily concentrated on adolescents or adults, with no direct investigation into the association between different regulators of IER and children's psychological well-being during middle and late childhood. The time when children are in school but prior to adolescence, characterized by parents' role being less predominant than that in early childhood and social interactions being more complex, is a pivotal period in children's lives (Collins, 1984). Thus, the goal of the present study was to investigate the role of different regulators (i.e., parents and friends) in the IER process and their impact on children's psychological adjustment using a person-centered approach. Additionally, the study aimed to examine age-related differences (i.e., grade) and gender across subgroups. Based on the aforementioned theories and previous research, the following hypotheses were formulated:

Hypothesis 1: Regarding the patterns of IER with parents and friends, given previous person-centered studies about social support and attachment, it was hypothesized that at least four profiles would emerge: (a) a profile characterized by low levels of IER with parents and high levels of IER with friends, (b) a profile characterized by high levels of IER with parents and low levels of IER with friends, (c) a profile with high levels of IER with both parents and friends, and (d) a profile characterized by low levels of IER with both parents and friends.

Hypothesis 2: Regarding the relationship between different IER patterns and children's psychological adjustment, we hypothesized that the profile demonstrating high levels of IER with both parents and friends would exhibit the best psychological adjustment compared to other profiles. Besides, it was hypothesized that children in the profile with low levels of IER with both parents and friends were expected to report the lowest levels of psychological adjustment. However, no particular hypotheses were established for the profiles characterized by different levels of IER with parents and friends because these analyses were exploratory. Moreover, due to the developing self-regulatory capacities of children (Gee et al., 2014; Steinberg et al., 2018), children in the lower grades may rely heavily on others to regulate emotions. It was also hypothesized that children in the profile with high levels of IER with both parents and friends may benefit more in the low grade with better adjustment.

Hypothesis 3: In addition to testing our main hypotheses, we also aimed to reveal the grade and gender characteristics within different profiles to gain a deeper insight into IER patterns. As previously stated, it was expected that children in the high grade or girls were more likely to be in the profile characterized by high levels of IER with friends in contrast to other profiles.

Method

Participants

Participants in this study were $N = 1{,}678$ children (822 girls and 854 boys, two did not specify their gender) with ages ranging from 8 to 13 years ($M_{\text{age}} = 10.42$, $SD = 1.16$) in Shandong Province, People's Republic of China. The children were randomly selected from two elementary schools across grade levels: Grade 3 ($n = 370$), Grade 4 ($n = 397$), Grade 5 ($n = 483$), and Grade 6 ($n = 428$).

In terms of parental education, approximately 90% of the mothers and fathers had completed high school or below, while the remaining parents had attained professional training, college, or university education. The majority of the families were from a middle socioeconomic status, with about half of the fathers and mothers earning a monthly income exceeding CN¥5,000 (about 698 U.S. dollars). Thirty-four percent of the children reported being the only child. Almost all children and parents were of Han ethnicity, the predominant ethnic nationality in China. Preliminary analyses revealed that these demographic variables did not significantly relate to the study's variables or relationships of interest.

Procedure

Institutional review board approval from East China Normal University was obtained before recruitment and data collection. Before data collection began, approval was obtained from participating elementary schools as well as informed consent from parents through the online platform, and finally, 90.9% of parents agreed to participate. Written informed consent was obtained from children of families who agreed to participate and ensured that the children participated in this study voluntarily. Assessments were administered by a group of trained research assistants (undergraduate and postgraduate psychology students) and conducted collectively during school hours in the classroom. Throughout the whole process, the children filled out the questionnaires independently without chatting with others, and research assistants would interpret the items if the children had any questions. Upon completion of the study, each child received a gift as compensation for their participation.

Measures

IER With Parents and Friends

We revised the original Interpersonal Regulation Questionnaire (IRQ; Williams et al., 2018) to assess children's IER with parents and friends using a 7-point Likert-type scale, ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). The questionnaire consisted of four factors: negative emotion IER tendency, negative emotion IER efficacy, positive emotion IER tendency, and positive emotion IER efficacy. It measures both the tendency to use IER and the perceived efficacy of IER in decreasing negative emotions and increasing positive emotions. Mean scores across the 16 items were calculated, with higher scores reflecting higher levels of IER. The original measure demonstrated good reliability and validity in the Chinese cultural context (Ding et al., 2021). For this study, we adapted the questionnaire by replacing the general term "others" with "parents" (e.g., I manage my emotions by expressing them to parents) and "friends" (e.g., I appreciate having friends' support through difficult times) in each item, resulting in two separate scales for IER with parents and friends. Confirmatory factor analysis confirmed the adequacy of the four-factor model for both scales: IER with parents ($\chi^2/df = 2.89$, comparative fit index [CFI] = 0.97, Tucker–Lewis index [TLI] = 0.97, root-mean-square error of approximation [RMSEA] = 0.03) and IER with friends ($\chi^2/df = 4.03$, CFI = 0.96, TLI = 0.95, RMSEA = 0.04). In this study, the McDonald's ω for the parents' scale was 0.90 (0.61, 0.67, 0.84, and 0.70 for four dimensions, respectively). For the friends' scale, the McDonald's ω was 0.91 (0.65, 0.70, 0.87, and 0.65 for four dimensions, respectively).

Depressive Symptoms

Children's depressive symptoms were assessed using the Chinese version of the Childhood Depression Inventory (Kovacs, 1992; revised by Chen et al., 2005). The Childhood Depression Inventory consisted of 14 items, each of which provided three alternatives from which the children selected the one that best matched their experiences over the past 2 weeks. The scale contained descriptions related to depressed mood as well as behavior (e.g., feeling lonely, self-blame, reduced social interest, anhedonia, reduced appetite).

Mean scores were calculated, with higher scores indicating more severe depression symptoms. The measure has demonstrated good reliability and validity among Chinese children (e.g., Hu et al., 2022). In the present study, the McDonald's ω was 0.87.

Anxiety Symptoms

Anxiety symptoms were measured using the self-rating anxiety scale developed by Zung (1971). This scale comprised 20 items, rated on a 4-point Likert scale (1 = *none or a little of the time* to 4 = *most or all of the time*). Mean scores were calculated, and higher scores indicate a higher frequency of anxiety symptoms. The scale has shown good reliability and validity among Chinese children (e.g., Xu et al., 2011), with a McDonald's ω of 0.78 in this study.

General Self-Worth

Children's general self-worth was assessed using the global self-perception subscale from the Self-Perception Profile for Children (Harter, 1985). This subscale consisted of six items. All items were rated on a 5-point scale from 1 (*completely disagree*) to 5 (*completely agree*; e.g., "I like myself"). Mean scores were calculated, with higher scores reflecting higher general self-worth. This measure has demonstrated good reliability and validity in Chinese samples (e.g., J. Liu et al., 2015). In the present study, the McDonald's ω was 0.86.

Life Satisfaction

Life satisfaction was measured with the Satisfaction With Life Scale (Diener et al., 1985). This scale consisted of five statements (e.g., "I am satisfied with my life"), with children rating their agreement to each statement on a 7-point Likert-type scale from 1 (*strongly disagree*) to 7 (*strongly agree*). Mean scores were calculated, with higher scores indicating greater life satisfaction. The scale has shown good reliability and validity in Chinese samples (e.g., Dou et al., 2023), and in the present study, the McDonald's ω was 0.78.

Analysis Plan

The rate of missing data ranged from 2.5% to 11.3% for the study variables and less than 2.4% missing values on the item level. Although Little's Missing Completely at Random test was significant, $\chi^2(854) = 956.210$, $p < .01$. However, the chi-square tests were overly sensitive to sample size (e.g., Kline, 2016; Ulman, 2013), and the normed chi-square (i.e., dividing the chi-square by the *df*) was applied (Kline, 2016). In light of previous studies (e.g., Arslan et al., 2023), $\chi^2/df = 1.12 < 3$, suggesting that the missing data can be considered to be missing completely at random. Missing data were handled using the full information maximum likelihood estimation (Graham, 2009) in *Mplus* 8.3.

Latent profile analysis (LPA) was employed to identify distinct profiles based on patterns of IER with different figures. To determine the optimal number of profiles, we evaluated several statistical indicators of model fit, including Akaike information criteria (AIC), Bayesian information criteria (BIC), sample-size-adjusted BIC (aBIC), and classification accuracy as measured by entropy. Lower values of AIC, BIC, and aBIC indicate a better fit, while higher values of entropy suggest a more accurate classification (Tein et al., 2013). Additionally, the bootstrap likelihood ratio test (BLRT) and the Lo–Mendell–Rubin likelihood ratio test (LMR) were used to compare models, with a significant p value indicating that a k-profile model was a significantly better fit than the $k - 1$ profile model (Muthén & Muthén, 2010). The BIC values and the BLRT p value were given special weight, as they are consistent indicators of class solutions (Nylund et al., 2007; Vermunt, 2010). The meaningfulness and interpretability of the profiles were also taken into account in selecting the final model.

Data analyses proceeded in three steps. First, descriptive statistics and repeated-measures analysis of variance were conducted. Second, a series of unconditional LPA models was analyzed, with the number of latent profiles ranging from two to six. The optimal number of profiles was determined based on the AIC, BIC, entropy, and the proportions of the profiles. Finally, multiple logistic regression and multivariate analysis of variance (MANOVA) were conducted to examine the associations between the latent profiles and predictors, as well as distal outcomes via SPSS 26.0. The Bonferroni correction was performed to adjust p values caused by multiple comparisons across profiles. All LPA was conducted in *Mplus* 8.3 software with the maximum likelihood robust estimator.

Transparency and Openness

The data set of this study and the *Mplus* script of the analyses have been available on the Open Science Framework at https://osf.io/839uv/?view_only=f8ad36c7704246889f9f0e01bd6ab085. This study's design and analysis were not preregistered.

Results

Preliminary Analyses

Repeated-measures analysis of variance was conducted on IER scores, with the regulators (parents vs. friends) as the within-subject factor and gender (boys vs. girls) and grade (low vs. high; low = Grades 3 and 4, high = Grades 5 and 6) as the between-subject factors. Results revealed a significant main effect of regulators, $F(1, 1413) = 20.54$, $p < .001$, $\eta_p^2 = .014$. Follow-up post hoc tests showed that children engaged in more IER with parents ($M = 5.51$, $SE = 1.10$) compared to friends ($M = 5.40$, $SE = 1.11$). No significant main effects were found for gender or grade ($ps > .05$). Additionally, two significant interaction effects emerged. First, there was a significant interaction between the regulators and gender, $F(1, 1413) = 5.36$, $p = .021$, $\eta_p^2 = .004$. Simple effects analysis revealed that boys used more IER with parents than with friends, whereas this difference was not significant for girls. Second, the interaction between the regulators and grade was also significant, $F(1, 1413) = 18.37$, $p < .001$, $\eta_p^2 = .013$. Children in lower grades used more IER with parents than with friends, while this difference was not significant in higher grades. The interaction between gender and grade was not significant, $F(1, 1413) = 2.46$, $p = .117$, nor was the three-way interaction between gender, grade, and regulator, $F(1, 1413) = 0.05$, $p = .816$.

Descriptive statistics and correlations among the study variables are presented in Table 1. IER with parents and friends was negatively correlated with children's depressive and anxiety symptoms

Table 1
Descriptive Statistics and Intercorrelations Study Variables (N = 1,678)

Study variable	*M* (*SD*)	1	2	3	4	5	6	Skewness	Kurtosis
1. IRQ with parents	5.51 (1.08)	—						−1.07	1.58
2. IRQ with friends	5.38 (1.12)	.57**	—					−0.86	1.05
3. Depression	1.36 (0.35)	−.45**	−.32**	—				1.30	1.58
4. Anxiety	1.66 (0.39)	−.34**	−.27**	.66**	—			0.85	0.95
5. General self-worth	3.96 (0.84)	.43**	.36**	−.67**	−.49**	—		−0.85	0.45
6. Life satisfaction	5.15 (1.24)	.51**	.36**	−.57**	−.45**	.61**	—	−0.48	−0.21

Note. IRQ = Interpersonal emotion regulation.
** $p < .01$.

and positively correlated with children's general self-worth and life satisfaction.

Identifying Intrinsic IER Patterns

Multiple LPA models were conducted using the eight indicator variables (see Table 2 for the model fit indices and profile proportions). Six distinct subgroups were ultimately identified. As the number of profiles increased, the information criteria (IC) values, including AIC and BIC, showed a gradual decrease. However, the BIC and sample-size-adjusted BIC (aBIC) levels stabilized after the five-profile model (see Figures 1 and 2). The Lo–Mendell–Rubin likelihood ratio tests revealed that the five-profile model provided a better fit than the four-profile model, with only marginal improvement observed when comparing the five-profile model to the six-profile model. The entropy for the six-profile model was higher than that for the five-profile model, with average latent class probabilities exceeding .85. Considering both the theoretical interpretability and meaningfulness of the models (e.g., Chui & Liu, 2021), the six-profile solution was chosen as the final model.

The mean values of the indicators based on the four factors of IER with parents or friends for each profile are shown in Table 3. Profile 1, labeled "extremely low IER" ($n = 42$, 2.5%), was characterized by the lowest levels of IER with parents and friends, scoring approximately 2 standard deviations (*SD*) below the average; Profile 2, labeled "low IER" ($n = 271$, 16.2%), was characterized by low levels of IER with parents and friends, with scores around half an *SD* below the average; Profile 3, labeled "average IER" ($n = 675$, 40.2%), was characterized by average levels of IER with parents and friends; Profile 4, labeled "high parent-low friend IER" ($n = 71$, 4.2%), was characterized by higher levels of IER with parents and lower levels of IER with friends; Profile 5, labeled "low parent-high friend IER" ($n = 65$, 3.9%), was characterized by lower levels of IER with parents and higher levels of IER with friends; Profile 6, labeled "high IER" ($n = 554$, 33.0%), was characterized by the highest levels of IER with parents as well as friends. These profiles are depicted in Figure 3.

Grade and Gender on Latent Profile Membership

The results of multinomial logistic regression were presented in Table 4, with gender, grade, and their interaction included simultaneously as independent variables on latent profile membership. For the predicting role of grade, children in higher grades were significantly more likely to belong to the low parent-high friend IER profile compared to all other profiles (*p*s < .05). Moreover, high graders were also more likely to be classified into average IER and high IER profiles in contrast to the extremely low IER profile. For the predicting role of gender, girls were more likely than boys to be classified into low parent-high friend IER profile compared to the extremely low IER, low IER, and high parent-low friend IER profiles (*p*s < .05).

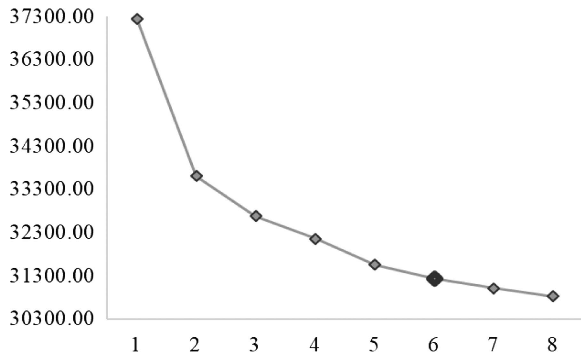
Moreover, the interaction between grade and gender significantly differentiated the children in the extremely low IER profile and all other profiles (*p*s < .05) except for the high parent-low friend IER profile. These interactions were presented in Figure 4. The interaction plot showed that as grade increased, boys were more likely to be in the other profiles than in the extremely low IER profile. However, compared to being in the extremely low IER profile, girls in higher grades were less likely to be classified into the low IER, average IER, and high IER profiles, while more likely to be classified into the low parent-high friend IER profile.

Table 2
Model Fit Indices for the Latent Profile Classification Analyses (N = 1,678)

Profile	AIC	BIC	SSA-BIC	LMR *p*	BLRT *p*	Entropy	Average latent class probability	Proportion per profile %
1	37160.46	37247.26	37196.43				1	
2	33472.61	33608.24	33528.82	.00	<.001	0.88	0.941–0.974	29.56/70.44
3	32496.17	32680.63	32572.62	.12	<.001	0.88	0.923–0.959	4.53/34.98/60.49
4	31919.71	32153.00	32016.39	.03	<.001	0.81	0.863–0.974	3.40/20.38/44.70/31.53
5	31269.21	31551.32	31386.13	.02	<.001	0.85	0.878–0.957	2.56/19.19/40.83/32.78/4.65
6	**30901.71**	**31232.65**	**31038.87**	**.50**	**<.001**	**0.86**	**0.867–0.947**	**2.50/16.15/40.23/4.23/3.87/33.02**
7	30631.20	31010.97	30788.59	.12	<.001	0.87	0.867–0.957	2.27/37.31/16.81/1.49/5.13/3.75/33.25
8	30389.74	30818.34	30567.37	.76	<.001	0.84	0.782–0.971	2.09/15.20/33.02/24.08/2.32/6.14/1.49/15.67

Note. Values presented in bold indicate the final selected profile. AIC = Akaike information criteria; BIC = Bayesian information criteria; SSA-BIC = sample-size-adjusted BIC; LMR $p = p$ value of Lo–Mendell–Rubin likelihood ratio test; BLRT $p = p$ value of the bootstrap likelihood ratio test.

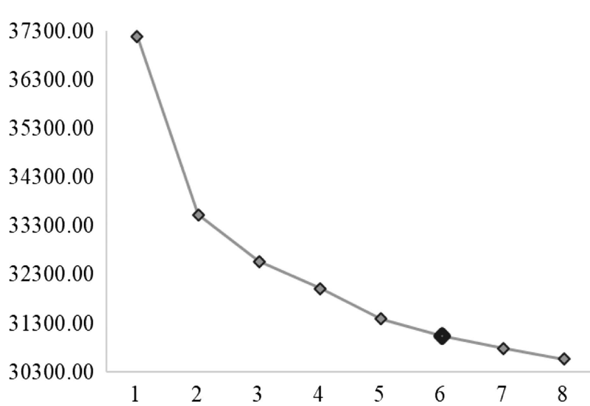
Figure 1
The Steep Slope of Bayesian Information Criteria



Intrinsic IER Profiles and Child Psychological Adjustment

A multivariate analysis of variance (MANOVA) was conducted with IER profiles and grade as independent variables and child psychological adjustment (i.e., depression, anxiety, general self-worth, and life satisfaction) as dependent variables. Results revealed the main effect of IER profiles was significant, $F(20, 4439) = 22.00$, Wilk's $\lambda = 0.73$, $p < .001$, $\eta_p^2 = .08$. Follow-up univariate analyses revealed significant differences among IER profiles on all adjustment indicators, with the partial eta-squared (η^2) effect sizes ranging from .10 to .20. Table 5 presents the pairwise comparisons of psychological adjustment outcomes across IER latent profiles. Children in the high IER profiles had the lowest levels of depressive and anxiety symptoms and the highest levels of general self-worth and life satisfaction, followed by those in the average IER profile and high parent–low friend IER profile. No significant differences between the average IER profile and the high parent–low friend IER profile were found across the adjustment indicators. Children in the remaining profiles generally reported poorer psychological adjustment compared to those profiles. Specifically, children in the remaining profiles all exhibited higher levels of depressive symptoms and lower levels of life satisfaction compared to the high parent–low friend IER profile. Moreover, similar to those in the extremely low IER profile, children who were in the low parent–high friend IER profile reported the highest level of depression and anxiety symptoms as well as the lowest level of general self-worth and life satisfaction. In addition, these children in the low parent–high friend IER profile also reported similar lower levels of adjustment as those of the low IER profile compared with the average IER and high IER profiles. The standardized estimates of psychological adjustment indicators across profiles are presented in Figure 5. Moreover, the main effect of grade was not significant ($p =$ .261). Though the interaction between IER profile and grade was significant, $F(20, 4439) = 1.66$, Wilk's $\lambda = 0.98$, $p = .033$, $\eta_p^2 = .01$, follow-up univariate analyses did not reveal significant differences across adjustment indicators (ps $> .05$).

Figure 2
The Steep Slope of Sample-Size-Adjusted Bayesian Information Criteria



Sensitive Analysis

To investigate the potential influence of missing data on IER items, we performed the LPA analysis based on the complete data sets of IER items ($n = 1{,}417$) and finally identified six profiles. The predicting roles of gender and grade on latent membership remain significant. For the interaction term between gender and grade, it was only found that boys as grade increased remained to be more likely to be in the low parent–high friend IER profile than in the extremely low IER profile. Moreover, the association between IER profile and children's psychological adjustment was also examined. Similar to the extremely low IER and low IER profiles, children in the low parent–high friend IER profile still exhibited lower levels of psychological adjustment compared to those in the average IER and high IER profiles. Besides, these children still reported higher levels of depressive symptoms and lower levels of life satisfaction compared with the high parent–low friend IER profile. Altogether, the main results remained significant, indicating the robustness of the current findings.

Discussion

The present study aimed to identify patterns of intrinsic IER among Chinese elementary school students using a person-centered approach and to examine how these patterns relate to children's psychological adjustment. To the best of our knowledge, this is the first study to identify distinct intrinsic IER profiles, describe the gender and age characteristics of these profiles, and explore their differential associations with children's psychological adjustment. Consequently, six meaningful profiles were identified, and grade or gender also played a predictive role across profiles. Children in the high IER profile exhibit the best psychological adjustment, while those in the extremely low IER, low parent–high friend IER, and low IER profiles all reported poorer adjustment.

Identification of Intrinsic IER Profiles and Demographic Predictors

The present study examined children's interpersonal emotional regulation with both parents and friends. The results revealed four profiles based on the levels of IER with these two types of regulators: extremely low, low, average, and high levels of IER with both parents and friends. Additionally, two mixed profiles characterized by different levels of IER with parents and friends were

Table 3
Results of Descriptive Data for Each Group and Post Hoc Comparisons (MANOVA)

Profile	NTP	NEP	PTP	PEP	NTF	NEF	PTF	PEF
Extremely low IER	−1.65 (0.61)$_a$	−2.68 (1.16)$_a$	−2.81 (0.99)$_a$	−2.16 (0.95)$_a$	−1.67 (0.96)$_a$	−2.78 (1.15)$_a$	−2.48 (0.81)$_a$	−2.19 (1.20)$_a$
Low IER	−0.73 (0.64)$_b$	−0.92 (0.78)$_c$	−0.99 (0.86)$_c$	−0.92 (0.72)$_b$	−0.83 (0.70)$_b$	−0.96 (0.65)$_c$	−1.05 (0.70)$_c$	−0.90 (0.78)$_c$
Average IER	−0.22 (0.77)$_c$	0.08 (0.60)$_d$	0.20 (0.63)$_d$	0.06 (0.79)$_c$	−0.20 (0.75)$_c$	0.04 (0.64)$_d$	0.12 (0.71)$_d$	0.05 (0.76)$_d$
High parent–low friend IER	0.35 (0.75)$_d$	0.24 (0.62)$_d$	0.39 (0.43)$_{d,e}$	0.14 (0.68)$_c$	−0.85 (0.78)$_b$	−1.51 (1.00)$_b$	−1.76 (0.83)$_b$	−1.65 (0.89)$_b$
Low parent–high friend IER	−1.45 (0.81)$_a$	−1.90 (1.15)$_b$	−1.77 (1.28)$_b$	−1.72 (1.05)$_a$	0.21 (1.14)$_d$	0.40 (0.64)$_e$	0.36 (0.71)$_d$	0.49 (0.67)$_e$
High IER	0.86 (0.68)$_e$	0.72 (0.46)$_e$	0.60 (0.34)$_e$	0.70 (0.48)$_d$	0.88 (0.63)$_e$	0.77 (0.45)$_f$	0.70 (0.41)$_e$	0.69 (0.49)$_e$

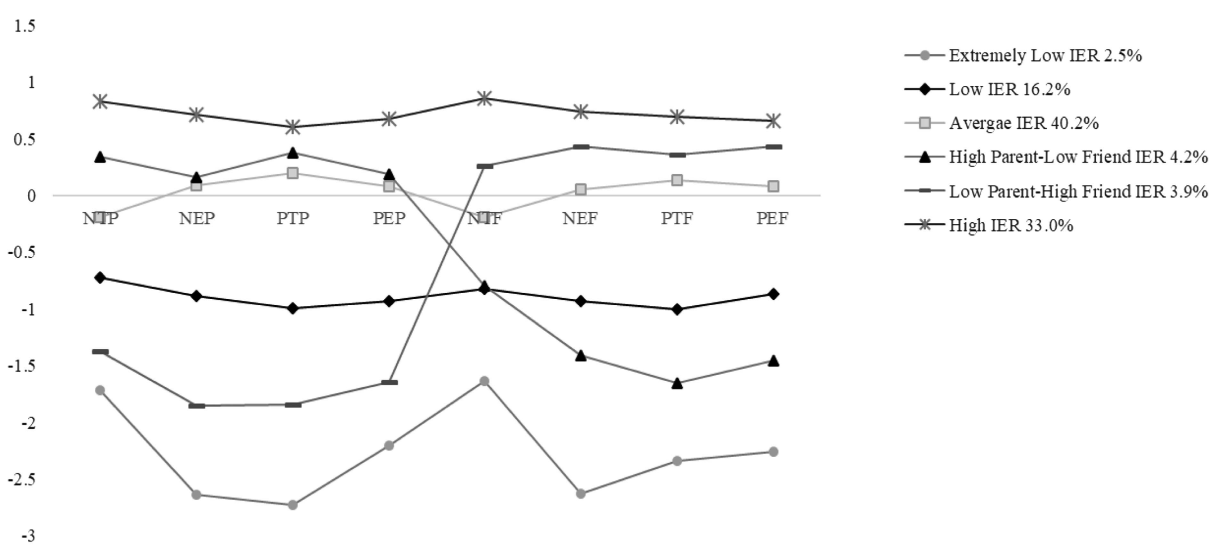
Note. The values in parentheses indicate standard deviations. Different subscripts in the same column indicate significant differences from one another ($p < .05$). a < b < c < d < e < f. MANOVA = multivariate analysis of variance; IER = interpersonal emotion regulation; NTP = negative emotion IER tendency with parents; NEP = negative emotion IER efficacy with parents; PTP = positive emotion IER tendency with parents; PEP = positive emotion IER efficacy with parents; NTF = negative emotion IER tendency with friends; NEF = negative emotion IER efficacy with friends; PTF = positive emotion IER tendency with friends; PEF = positive emotion IER efficacy with friends.

also identified, which is consistent with findings from previous research (e.g., Laible et al., 2000; Petersen et al., 2023). In line with attachment theory (Bowlby, 1982), internal working models formulated from attachments to parents would shape subsequent social relationships, such as friendship. In other words, children who rely heavily on parents to manage their emotions are also more prone to seek help from friends. Conversely, children who show less dependency on parents would be less prone to seek help from friends to regulate emotions. Thus, four profiles characterized by the different levels of IER with both parents and friends were identified. It is noteworthy that the high IER profile accounted for 33.0% of the sample, underscoring the significant role of social interaction in emotion regulation. As children enter middle childhood, their interactions with peers increase substantially compared to earlier years (e.g., Lam et al., 2014; Rubin et al., 2006). Consequently, a meaningful profile characterized as high IER with friends and low IER with parents has emerged. Moreover, children also treat their parents as primary attachment figures, so a profile was labeled low IER with friends and high IER with parents.

The present study found that children in the higher grade were more likely to belong to the low parent–high friend IER profile compared to other profiles generally. This suggests that as children progress through grades, friends increasingly play a significant role in their lives. This finding was consistent with previous research indicating that older children tend to trust and communicate more with their friends due to increased peer interactions (Nickerson & Nagle, 2005). Moreover, it was also found that they were more likely to be in the average IER and high IER profiles in contrast with the extremely low IER profile, indicating children are actively engaging in IER with others as their relationships become more diverse (Barthel et al., 2018).

In terms of gender differences across profiles, as expected, girls were more likely to be in the low parent–high friend IER profile

Figure 3
Standardized Means of Indicators Presented by Profiles



Note. NTP = negative emotion IER tendency with parents; IER = interpersonal emotion regulation; NEP = negative emotion IER efficacy with parents; PTP = positive emotion IER tendency with parents; PEP = positive emotion IER efficacy with parents; NTF = negative emotion IER tendency with friends; NEF = negative emotion IER efficacy with friends; PTF = positive emotion IER tendency with friends; PEF = positive emotion IER efficacy with friends.

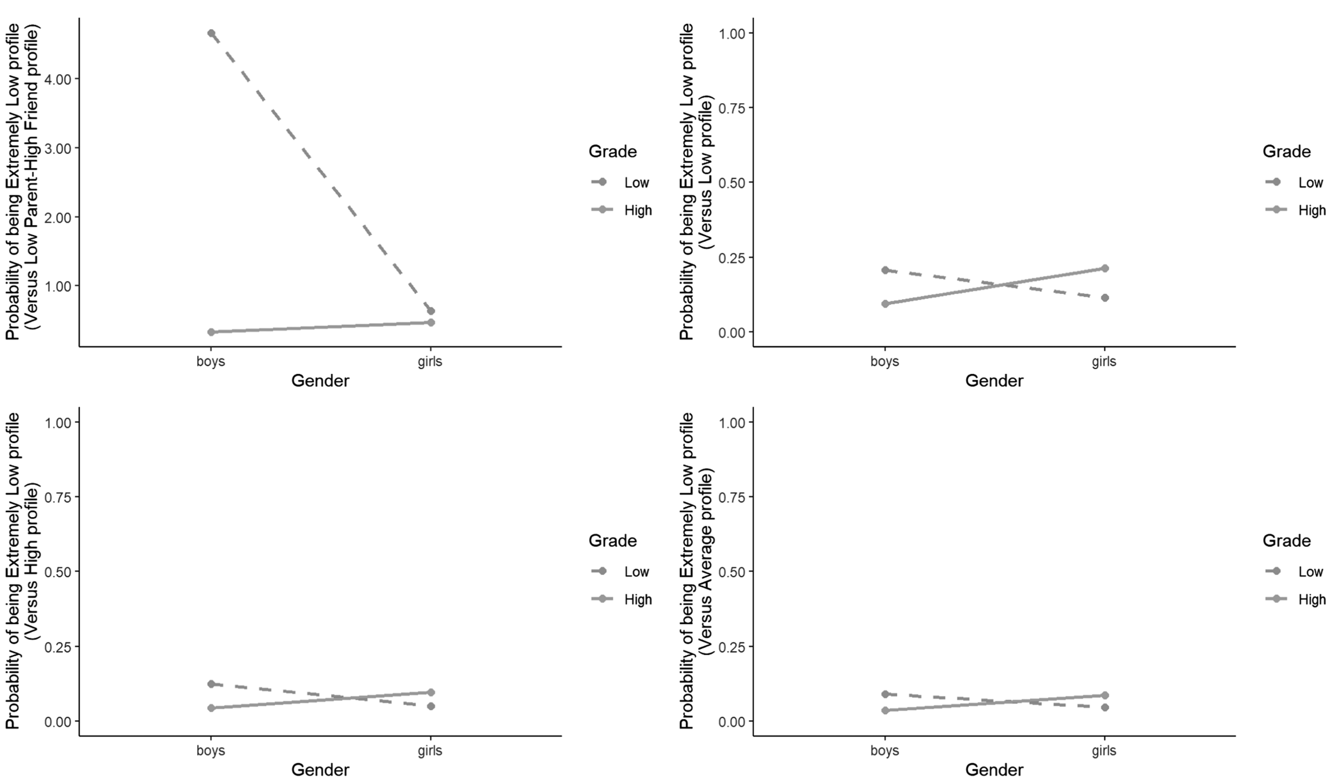
Table 4
Gender, Grade, and the Interaction Between Two Predicating Profiles

Comparison	Reference	Gender (girls vs. boys) Logit (*SE*)	*OR*	Grade (high vs. low) Logit (*SE*)	*OR*	Gender × Grade (Girls-High vs. Others) Logit (*SE*)	*OR*
Extremely low IER	Low parent–high friend IER	**−1.99 (0.80)***	0.14	**−2.64 (0.77)****	0.07	**2.33 (0.97)***	10.27
Low IER		**−1.41 (0.67)***	0.24	**−1.85 (0.64)****	0.16	0.92 (0.75)	2.52
Average IER		−1.29 (0.66)	0.27	**−1.69 (0.63)****	0.19	0.72 (0.73)	2.06
High parent–low friend IER		**−1.60 (0.73)***	0.20	**−2.25 (0.70)****	0.11	1.05 (0.86)	2.86
High IER		−1.09 (0.66)	0.34	**−1.60 (0.63)***	0.20	0.65 (0.73)	1.93
Low IER	Extremely low IER	0.59 (0.50)	1.79	0.79 (0.49)	2.21	**−1.41 (0.70)***	0.25
Average IER		0.70 (0.48)	2.01	**0.95 (0.48)***	2.59	**−1.61 (0.67)***	0.20
High parent–low friend IER		0.39 (0.56)	1.48	0.39 (0.56)	1.48	−1.28 (0.81)	0.28
High IER		0.90 (0.48)	2.46	**1.04 (0.48)***	2.82	**−1.67 (0.68)***	0.19
Average IER	Low IER	0.12 (0.21)	1.12	0.16 (0.20)	1.18	−0.20 (0.29)	0.82
High parent–low friend IER		−0.19 (0.37)	0.82	−0.40 (0.36)	0.67	0.13 (0.54)	1.14
High IER		0.32 (0.22)	1.37	0.25 (0.21)	1.28	−0.27 (0.30)	0.76
High parent–low friend IER	Average IER	−0.31 (0.34)	0.73	−0.56 (0.34)	0.57	0.33 (0.51)	1.39
High IER		0.20 (0.17)	1.22	0.08 (0.16)	1.09	−0.07 (0.23)	0.94
High IER	High parent–low friend IER	0.51 (0.34)	1.67	0.65 (0.34)	1.91	−0.40 (0.51)	0.67

Note. Significant differences were marked in bold. *SE* = standard error; IER = interpersonal emotion regulation. Grades 3 and 4 and Grades 5 and 6 were classified into low and high grades, respectively.
* $p < .05$. ** $p < .01$.

compared to the extremely low IER, low IER, and high parent–low friend IER profiles. This is consistent with the relational orientation style commonly observed in girls, who tend to engage more in social interactions as they reach middle childhood (Ferrar et al., 2012; Ladd, 1983; Moller et al., 1992). Additionally, girls are found to engage in greater self-disclosure with their friends than boys, suggesting that they may be more inclined to seek help from their friends for effective emotion regulation (Rose & Rudolph, 2006).

Figure 4
Interaction Effects From the Multinomial Logistic Regression Analysis Between Gender and Grade on Probability of Interpersonal Emotion Regulation's Membership



Note. The *y*-axis range of the first figure is different from that of the other three figures. IER = interpersonal emotion regulation.

Table 5
Differences Between Profiles on Standardized Means of Psychological Adjustment

Outcome	Extremely low IER	Low IER	Average IER	High parent-low friend IER	Low parent-high friend IER	High IER
Depression	1.02 (1.49)$_c$	0.56 (1.17)$_c$	−0.09 (0.85)$_b$	0.16 (0.92)$_b$	0.86 (1.16)$_c$	−0.45 (0.74)$_a$
Anxiety	1.05 (1.32)$_d$	0.46 (1.02)$_c$	−0.06 (0.94)$_b$	0.27 (1.01)$_{b,c}$	0.65 (1.05)$_{c,d}$	−0.30 (0.87)$_a$
General self-worth	−0.84 (1.35)$_a$	−0.61 (0.93)$_a$	−0.02 (0.90)$_c$	−0.20 (1.18)$_{b,c}$	−0.58 (0.93)$_{a,b}$	0.48 (0.79)$_d$
Life satisfaction	−1.06 (1.24)$_a$	−0.64 (0.89)$_a$	−0.08 (0.89)$_b$	−0.09 (1.02)$_b$	−0.70 (0.98)$_a$	0.57 (0.78)$_c$

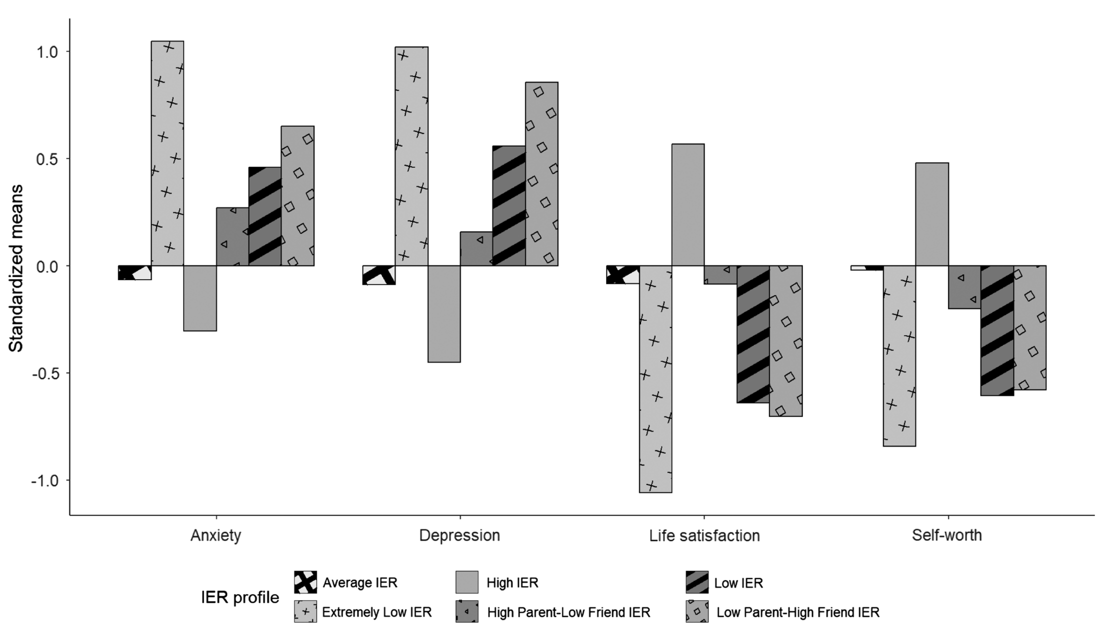
Note. Different subscripts in the same column indicate significant differences from one another ($p < .05$). a < b < c < d. The values in parentheses indicate standard deviations. IER = interpersonal emotion regulation.

Moreover, the results also revealed that as they progressed to a higher grade, compared with girls, boys were more likely to be in the low parent–high friend IER profile than in the extremely low IER profile. In other words, the switch to seeking friends for regulating emotions may be more salient among boys as grades increase. However, it should be noted that girls were more likely to be in the low parent–high friend IER profile than in the extremely low IER profile regardless of the grade level (see the probability below 1 in Figure 4). Additionally, when moving up to the upper grade, girls were found to be less likely to be in the low IER, average IER, and high IER profiles than in the extremely low IER profile, while not for boys. In other words, the likelihood of being in the extremely low IER profile among girls may increase in the high grade. This finding may reflect gender-specific characteristics during the transition to adolescence. Girls typically experience earlier pubertal maturation than boys and may begin to navigate physical and emotional changes independently (Negriff & Susman, 2011; Thornburg, 1980). This emerging desire for independence could reduce their tendency to seek external support for emotion regulation. However, this result was not robust in the sensitive analysis, which needs to be interpreted with caution.

Relation Between Intrinsic IER Profiles and Psychological Adjustment

The results from the present study showed that the high IER profile exhibited better psychological adjustment compared with other profiles, regardless of the grade level, highlighting the essential role of IER in reducing distress (e.g., Levy-Gigi & Shamay-Tsoory, 2017; Marroquín, 2011). While this result was consistent with previous studies (e.g., Laible et al., 2000; Petersen et al., 2023), our study specifically focused on middle and late childhood, a crucial yet underexplored transitional period leading to adolescence. In line with the relational regulation theory (Lakey & Orehek, 2011), children in the high IER profile benefit from a broader diversity of relationships and greater social support, or they select specific regulators for particular emotions. This increases the likelihood of effective regulation when faced with stressful events compared to other IER profiles (Cheung et al., 2015). Moreover, it was found that engaging in intrinsic IER strategies may be more beneficial for people in East Asian countries (e.g., Chinese) with a greater emphasis on interdependence with others than those in Western countries (Liddell & Williams, 2019; Yang & Maccann, 2024). Thus, the adaptiveness of intrinsic IER may be more

Figure 5
Standardized Means of Psychological Adjustment Across Profiles



Note. IER = interpersonal emotion regulation.

salient among children in China, which also accounts for the extremely low IER and low IER profiles with poorer adjustment.

Surprisingly, children in the low parent–high friend IER, extremely low IER, and low IER profiles all reported higher levels of depressive symptoms and anxiety symptoms and lower levels of general self-worth and life satisfaction compared to the average IER and high IER profiles. Notably, there were no significant differences in adjustment outcomes between the low parent–high friend IER and extremely low IER profiles. IER could be regarded as a mechanism of social support to some extent (Marroquín, 2011). This result was similar to the study focused on social support profiles among adolescents (Scholte et al., 2001), indicating that friends may not serve as an effective buffer for the children who used less IER with their parents, and children were biased to view them as supportive and effective. Several reasons could explain these findings. First, friends may provide limited external resources for effective emotion regulation. During middle childhood, children's self-regulatory behaviors are immature (Zachariou & Whitebread, 2019), and their control skills continue to develop after age 9 (Roebers et al., 2009). Although children may share their emotions and perceive the efficacy of regulating their emotions with friends, their friends may not be able to complete the regulation tasks successfully (e.g., identify the target's emotions, select the appropriate strategy, or implement the strategy) due to developing fluid intelligence and crystallized intelligence (Niven, 2022), as well as the cognitive ability (Niven et al., 2024). Second, we only measured the tendency and efficacy of IER, providing limited insight into the regulatory process during social interactions. If children engage in rumination with their friends, it could be associated with internalizing problems (Rose, 2002, 2021). Therefore, the dynamics of IER in children's social interactions warrant further investigation to fully understand these complex relationships.

Altogether, the results highlighted that parents play a critical role in children's adjustment during the interpersonal emotion regulatory process, which was consistent with previous studies in related domains such as attachment and social support (Armsden et al., 1990; Helsen et al., 2000). According to the social baseline theory, a healthy early family environment provides essential external resources for the social baseline of the human brain (Beckes & Coan, 2011). This secure base is crucial for children to regulate their emotions as they grow older (Bowlby, 1982; Brumariu, 2015). In other words, children rely heavily on their parents' practice for emotion regulation during early life and then gradually develop distributed relationships that contribute to this regulation (Barthel et al., 2018; Hofmann & Doan, 2018; Morris et al., 2007). This foundational support from parents allows children to cultivate emotional regulation skills that are reinforced and diversified through interactions with peers and other significant figures in their lives.

Limitations and Future Directions

The present study provided insights into the heterogeneous IER patterns of how children regulate themselves through interacting with parents or friends. Nevertheless, the present study has several limitations. First, we focused solely on the parents and friends as the regulators. As children grow, their social networks become more complex, involving other significant figures such as teachers, siblings, and grandparents (Barthel et al., 2018). Future research should include these additional relationships to provide a more comprehensive understanding of IER.

Second, the cross-sectional nature of the data limited our ability to reveal changes and stability between the different profiles, as well as the factors that may predict the transition. Moreover, while we concluded that high IER is associated with better psychological adjustment, it is also possible that children with good mental health are more inclined to use IER with others. Future studies should explore these potential bidirectional relationships using longitudinal designs to better understand the dynamics between IER and adjustment.

Third, the present study investigated the relationship between intrinsic IER and children's adjustment but ignored how the regulators manage their emotions. Moreover, the tendency to apply IER and children's perceived efficacy of intrinsic IER assessed in the present study was relatively high (see mean scores in Table 1), which limits the exploration of potential factors explaining variances in IER (e.g., small effect sizes about regulators). Thus, future studies should also consider assessing the extrinsic IER (e.g., how the children's parents or friends manage their emotions) simultaneously. Understanding these processes may provide insight into why the low parent-high friend IER profile was related to maladjustment as well.

Finally, the generalizability of results to other countries or regions remained uncertain, given the specific cultural context of our sample. Moreover, the small percentages of group sizes (e.g., low parent–high friend IER and high parent–low friend IER profiles) less than the threshold of 5% also limit the generalizability to other samples (Ferguson et al., 2020), though these groups were interpretable and substantively meaningful. Additionally, from the developmental perspective, children's relationships become more complex with age (Barthel et al., 2018). Thus, follow-up studies should extend to other age groups and take other relationships into account (e.g., romantic relationships, work relationships) across different cultural backgrounds using a person-centered approach.

Implications

Despite these limitations, the present study also has several implications. Though Niven et al. (2012) took distinct types of relationships into account, little is known about how the intrinsic IER varies across relationships. To the best of our knowledge, the present study was the first to explore the tendency to use IER and their evaluations of its efficacy with different regulators among elementary children by adopting a person-centered approach to reveal more heterogeneous information, particularly the differential effects of interpersonal emotion regulators. Moreover, the findings also reflect the "risky" side of intrinsic IER from the person-centered perspective (Kwon et al., 2025; Zaki & Williams, 2013), since relying only on immature friends to handle emotions may increase the greater risk of regulation failure, which even intensifies their distress. Thus, parents should still provide enough emotional support for children in middle and late childhood, especially for girls. Besides, more attention and appropriate interventions should be given to children who refuse to seek help from others or only rely on friends to regulate emotions for clinicians and educators.

Conclusion

In conclusion, the present study highlighted the heterogeneity among children when they rely on different regulators to manage emotions. By utilizing a person-centered approach among Chinese pupils, we identified six distinct subtypes of intrinsic IER: extremely low IER, low IER, average IER, high parent–low friend IER, low parent–high friend IER, and high IER. Moreover, the present study revealed the predictive role of grade and gender on profiles, as well as the relationship between intrinsic IER profiles and psychological adjustment. It should be noteworthy that children who rely solely on friends for emotion regulation may be at greater risk for psychological maladjustment, as well as children with less help-seeking behavior from others. Collectively, the present study examining intrinsic IER patterns helps understand how children regulate themselves through interacting with parents or friends and underscores the continued importance of parents in emotion regulation during middle and late childhood.

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